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VORTEX PT

POST-TENSION CAD DETAILING & SHOP DRAWING AUTOMATION SUITE

Comprehensive Technical Guide for Automated Parabolic Profiling, RAM Concept & ADAPT-Builder Integration, Dual Hexagon Tendon Tagging, and Production AutoCAD DWG Generation.



CORE ENGINES	CAD STANDARDS	COMPLIANCE CODE	SECURITY & CLOUD
RAM Concept (.cpt) ADAPT (.inp / .inb)	AutoCAD DWG / DXF Excel Schedules	ACI 318-19 • PTI 6th Eurocode 2 EN 1992	Hardware-Locked PC AppData Roaming

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1. Introduction & Core Architecture

Vortex PT is a specialized desktop engineering suite designed to bridge the workflow gap between finite-element post-tensioned design software (**Bentley RAM Concept** and **ADAPT-Builder**) and construction-ready shop drawings in **AutoCAD (.dwg / .dxf)**.

Traditional shop drawing workflows require draftsmen to manually trace tendon spans, hand-calculate chair support heights, transcribe profile points, and manually draw anchor setbacks. Vortex PT eliminates these manual steps, reducing a multi-day drafting task to under 30 seconds while guaranteeing zero transcription errors.

TARGET USERS
Structural Engineers, PT Contractors, CAD Detailers

EXECUTION ENVIRONMENT
100% Offline Desktop, Windows 10 / 11 64-bit

INPUT FORMATS SUPPORTED
Bentley RAM Concept (*.cpt, *.cpt8) & ADAPT-Builder (*.adm with .inb/.inp)

OUTPUT FORMATS GENERATED
Layered AutoCAD DWG, DXF, Microsoft Excel (.xlsx), CSV

> **No Host Software License Required**
>
> Vortex PT features autonomous native binary parsers. You do **not** need a Bentley RAM Concept or ADAPT-Builder license running on your drafting or site workstations to open models and produce fabrication drawings.

02 Opening & Importing Models

2.1 Supported File Types

Vortex PT natively imports structural data directly from the industry's two dominant post-tensioned design platforms:

- **Bentley RAM Concept Files (`*.cpt`, `*.cpt8`)**: Direct parsing of proprietary binary archives including slabs, drop panels, openings, columns, shear walls, beams, latitude tendons, longitude tendons, and control points.
- **ADAPT-Builder Project Models (`*.adm` with companion `*.inb` / `*.inp`)**: Direct loading of ADAPT Floor Pro, MAT, and Edge models. You select the native `.adm` file directly in Vortex PT, which automatically reads the companion `.inb` / `.inp` file of the same name.

2.2 ADAPT-Builder Import Architecture (.adm + .inb)

In ADAPT-Builder, models are saved with the default extension **`.adm`**. To import an ADAPT model into Vortex PT:

- Vortex PT directly opens the **`.adm`** file from the file open dialog.
- **Mandatory Pairing Rule:** Beside your `.adm` file in the same folder, there must exist a companion Generic Model file (**`.inb`** or **`.inp`**) bearing the **exact same base name** (e.g. `Building_Level2.adm` paired with `Building_Level2.inb` or `Building_Level2.inp`).
- Vortex PT also automatically checks companion project subfolders (`<model_folder>/<model_name>/`) and ADAPT automatic backup directories (`backup/`).

> 💡 **How to Export from ADAPT-Builder**
>
> 1. Open your project in **ADAPT-Builder** (Floor Pro, MAT, or Edge).
> 2. Go to menu: **File → Export → Generic Model (*.inp / *.inb)**.
> 3. Save the exported file in the same directory as your `.adm` file with the matching name.
> 4. In Vortex PT, click **Open Model** and select the `.adm` file. Vortex PT reads the paired data and instantly renders the entire floor plate with 100% geometric fidelity!

2.3 Step-by-Step Model Loading Workflow

1. Click the vibrant electric cyan **"+ Open Model"** button located at the top left of the workspace toolbar, or press the universal shortcut `Ctrl` + `O`.
2. In the Windows File Dialog, browse to your model directory and select your **Bentley RAM Concept (`.cpt` / `.cpt8`)** or **ADAPT-Builder (`.adm`)** file.
3. Vortex PT initiates its multi-stage extraction pipeline:
   - Reads concrete geometry: slab perimeters, thickness zones, drop caps, and core openings.
   - Extracts structural supports: columns, shear walls, and perimeter foundation lines.

- Assembles continuous tendon strings from segmental FEA elements, calculating curvature, inflection transitions, and anchorage conditions.

4. Upon completion, the main CAD viewport renders the complete slab floor plate, and the **Project Metrics Card** populates immediately.



Figure 2.1: Live 2D CAD Workspace after opening `B01-PART 2.cpt`. Displays 70 continuous tendons, slab boundaries, drop panels, columns, the floating Project Metrics HUD, and Tendon Y-3 longitudinal profile.

2.4 ADAPT-Builder Exact Elongation & Area Extraction

When importing ADAPT-Builder models (`.inp` / `.inb`), Vortex PT guarantees 100% computational fidelity:

- **Exact Cross-Sectional Area (Aps)**: Preserves the exact total strand area (e.g. 197.4 mm² for 2 strands) as modeled in ADAPT, eliminating rounding discrepancies in force and stress calculations.
- **End Elongations (End 1 & End 2)**: Extracts theoretical jacking extensions for both stressing boundaries directly from ADAPT design output tables, matching fabrication tickets identically.

2.5 RAM Concept Transverse Beam & Flat Support Fidelity

Vortex PT incorporates intelligent geometry heuristics to handle complex post-tensioning conditions in Bentley RAM Concept (`.cpt`):

- **Transverse Beam Steps & Band Crossings**: Tendons crossing transverse band beams maintain true physical trajectory. Slices crossing linear ramp steps evaluate without spurious sag artifacts.

- **Elimination of False Low Sags**: Intermediate chairs between two high support zones strictly maintain high profile chair height (e.g. 155 mm) rather than dipping into artificial sags.
- **Slanted Edge Collinear Dims**: Non-orthogonal slab perimeters generate aligned dimension strings running parallel to the exterior concrete edge, preserving architectural boundaries.

03 2D CAD Viewport & Layer Controls

3.1 Viewport Navigation & Canvas Controls

The central viewport is powered by a high-performance vector rendering canvas designed for large multi-span commercial slabs:

ACTION	INPUT MECHANISM	DESCRIPTION
Pan Canvas	Middle Mouse Click & Drag or Right Mouse Click & Drag	Smoothly translates the viewport in any direction without altering zoom scale.
Zoom In / Out	Mouse Scroll Wheel	Continuously zooms centered directly onto the cursor coordinate.
Fit to Extents	Press `Spacebar` or click **"Fit View"**	Calculates the bounding box of all slabs and centers the entire floor plate.
Select Tendon	Left Mouse Click on Tendon	Selects the tendon, activates bright cyan vector highlight, and docks profile view.
Tendon Search	`Ctrl` + `F` → Type Tendon Mark	Instantly queries tendons (e.g. `X-14`, `Y-3`), highlights, and centers the view.

3.2 Layer Visibility Chips

Located in the command bar above the CAD canvas, interactive layer chips permit instant toggling of individual drafting layers:

- **Slabs (Gray #475569)**: Concrete floor boundaries and thickness elevation zones.
- **Drops (Orange #F59E0B)**: Drop panels and drop caps at column heads.
- **Columns (Yellow #FBBF24)**: Rectangular, circular, and wall support sections.
- **Beams (Cyan #06B6D4)**: Post-tensioned band beams and drop beams.
- **Tendons (Latitude Magenta #D946EF / Longitude Teal #14B8A6)**: Tendon c/c lines and anchor bubbles.
- **Chairs & Dimensions**: Chair height callouts, remainder dimensions, and span spacing.

3.3 Floating Project Metrics Card

The upper-right HUD card provides real-time engineering summary statistics for the loaded structure:

- **Slab Area & Thickness**: Net structural floor area (m²) and primary plate depth (mm).
- **Structural Members**: Drop panels count and transfer/band beams detected.
- **Tendon & Strand Length**: Total active tendon count and linear strand length (m).
- **Strand Weight Takeoff**: Total fabricated strand weight (kg) calculated using your configured unit weight.
- **Anchorage Breakdown**: Exact count of Live Stressing Jacks vs. Dead Blind End Anchors.

04 Parabolic Tendon Profiling & Chairs

4.1 The Longitudinal Profile Viewer

Selecting any tendon triggers the bottom **Longitudinal Tendon Profile Dock**. This viewer displays the continuous cross-sectional elevation of the tendon from its starting anchor to its ending anchor.



Figure 4.1: Continuous 11-Span Longitudinal Elevation Profile of Tendon X-16. Features high support points over columns, low mid-span sags with fabricated chair heights, pan box setback blocks, and span dimension strings.

4.2 Mathematical Profile Modeling & Inflection Points

Vortex PT calculates true multi-span second-degree and third-degree reverse parabolic curves across each support bay:

- **High Support Points (Hi)**: Peak elevations located over column centerlines or interior wall supports.
- **Low Sag Points (Li)**: Maximum sag elevations located at mid-spans to optimize balancing load.
- **Inflection Transition Points (Xinf)**: Mathematically computed points of zero curvature where the curve transitions smoothly between concave-up and concave-down parabolas.

4.3 Automatic Chair Support Calculations

Fabrication chairs are calculated at regular field intervals (default 1,000 mm) along the tendon:

Chair Height Governing Formula

`Fabricated Chair Height = Tendon Centerline Profile (CGS) − Deduction Constant`

Calculated chair heights are automatically rounded to your specified fabrication increment (e.g. 5 mm steps).

⚠ Anchor Placement vs. Chairs

Anchorage points (Live Jacking Ends and Dead Cast-In Anchors) are physical fittings affixed to the slab formwork. Vortex PT excludes anchors from the chair table summary to prevent accidental chair over-ordering on site.

05 Engineering Compliance & QA Alerts

5.1 Automated Quality Assurance Engine

Every imported model undergoes continuous geometric and structural integrity diagnostics based on **ACI 318-19** and the **Post-Tensioning Institute (PTI) Manual**. When an issue is detected, an amber **"⚠ QA Warning"** badge illuminates in the top command bar.

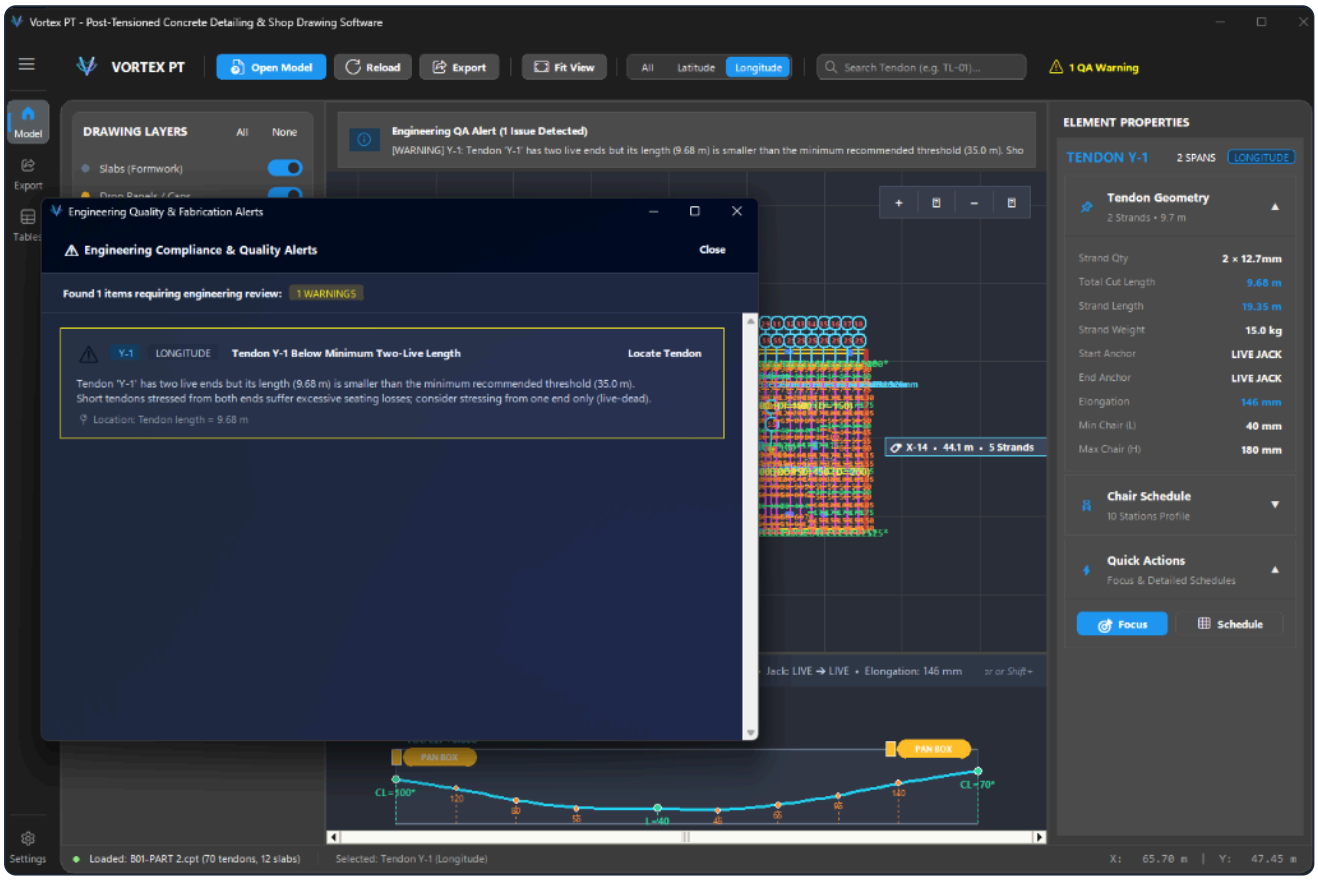


Figure 5.1: Engineering Compliance & QA Alerts diagnostic modal showing automated violation detection, severity ratings, and actionable structural recommendations.

5.2 Core Diagnostic Rules

A. Minimum Concrete Cover Verification

- **Top Cover Check**: Verifies that `Slab Depth − Profile Elevation ≥ Min Cover` (default: 25 mm).
- **Bottom Cover Check**: Verifies that `Profile Elevation ≥ Min Cover` (default: 25 mm).
- **Alert Condition**: If strand trajectory violates cover rules at high or low points, Vortex PT flags the exact bay and span coordinate.

B. Double Live End Minimum Length Threshold

- **Rule Rationale**: Short tendons configured with live stressing jacks at both ends lead to wedge seating loss issues and unnecessary labor costs.
- **Threshold Setting**: In **Settings**, you configure the warning threshold (e.g. 35.0 m). Tendons with two live ends shorter than this threshold trigger an alert advising a single live end configuration.

C. Two Dead Ends Critical Validation

- **Fatal Check**: Scans the model for any tendon terminated with dead-end (cast-in) anchors at both boundaries.
- **Severity**: High / Critical. Tendons cannot be stressed if both ends are encapsulated in blind anchors.

06 Detailing Settings & CAD Blocks

6.1 Windows 11 Fluent Settings Console

Click the **"Settings"** gear icon in the left navigation rail or press `Ctrl` + `,` to open the comprehensive configuration console.

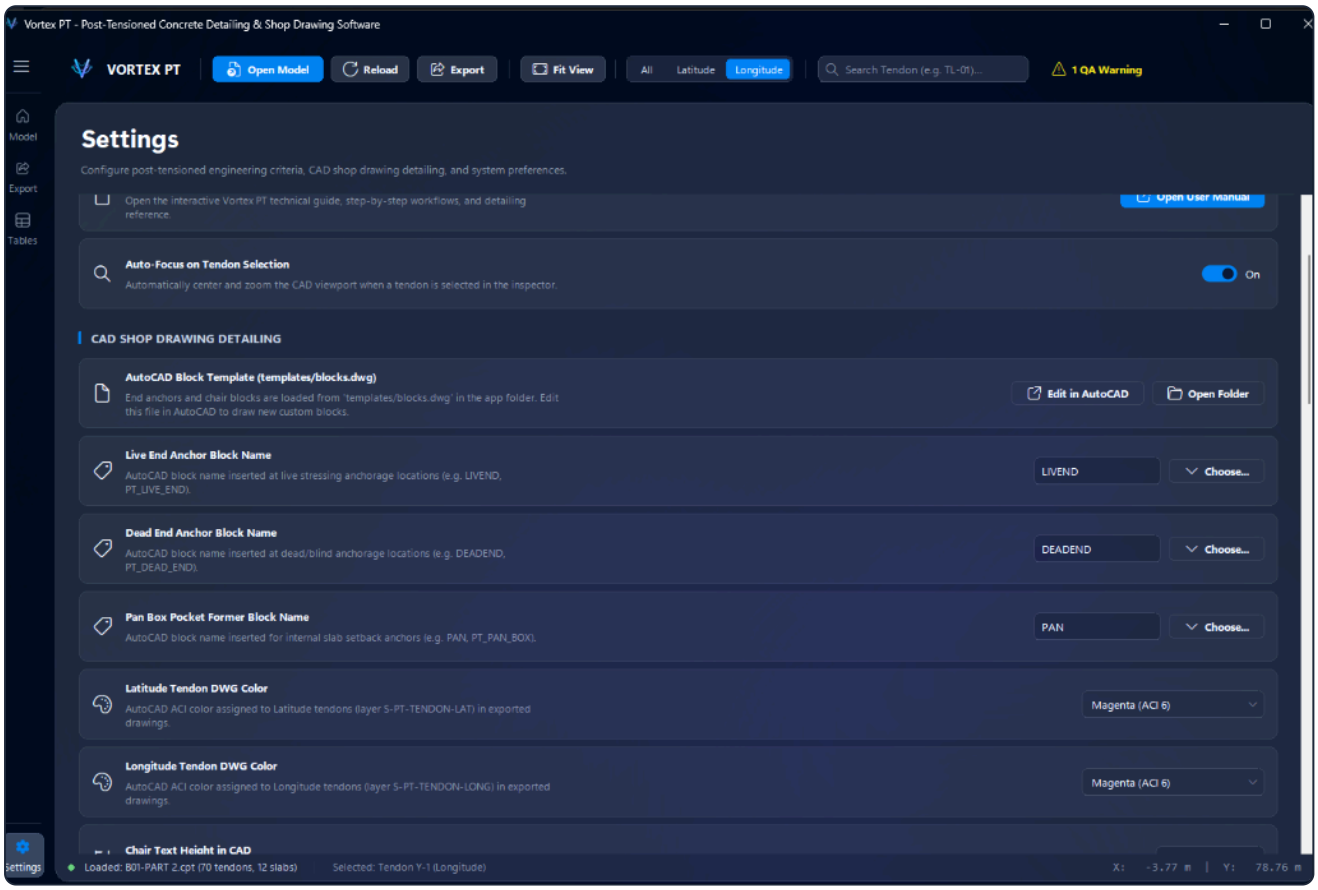


Figure 6.1: CAD Shop Drawing Detailing configuration cards, including template selection, block name bindings (LIVEND, DEADEND, PAN), ACI layer color overrides, and strand properties.

6.2 The New Tendon ID Block System (BLOQUE1 & BLOQUE1_L)

Vortex PT features an advanced, industry-standard dual-geometry Tendon ID block architecture that separates plan orientation from sequence numbers:

BLOCK GEOMETRY TAG	ATTRIBUTE TAG	DISPLAY / FUNCTION
Top Hexagon	`ID-PLAN`	Directional axis indicator (**X** for Latitude, **Y** for Longitude).
Bottom Hexagon	`ID-NUMBER`	Sequence number of the tendon within the floor plate (e.g. `01`, `02`, `18`).
Center Circle	`STRANDS`	Strand count fabricated in the tendon bundle (e.g. 2, 4, 5).
Invisible Metadata	`MARK`, `LENGTH`, `ELONGATION1`, `ELONGATION2`, `CUT-LENGTH`, `SLAB`	Embedded non-printing attributes for contractor data extraction, BIM audits, and automated BOM sync.

> 🎯 **Automatic Tendon Slope Alignment & Zero Mirroring**
>
> The Tendon ID block is automatically rotated parallel to the tendon slope vector. The engine enforces quadrant normalization so text is **never mirrored, inverted, or rendered upside down**, regardless of tendon angle or stressing direction.

6.3 Persistent User Templates & Roaming AppData Isolation

To ensure your drafting customizations are never lost when installing software updates, Vortex PT isolates active user templates from the application binary directory:

- **Active User Template Location**: `%APPDATA%\VortexPT\templates\blocks.dwg` and `blocks.dxf`.
- **Zero-Overwrite Protection**: Software updates install exclusively into `C:\Program Files\Vortex PT`. The installer **never touches or overwrites** your custom blocks or `settings.json` in AppData.
- **Dedicated Controls in Settings**:
  - **Edit in AutoCAD**: Directly launches your live AppData template in AutoCAD.
  - **Open Folder**: Opens the AppData templates folder in Windows Explorer.
  - **Reset Defaults**: Safely restores factory template blocks while creating an automatic timestamped backup of your current blocks.

6.4 Material & Dimension Properties

- **Chair Text Height**: Configures the annotation height of chair labels in AutoCAD (default: `200 mm`).
- **Strand Unit Weight**: Sets the specific weight per linear meter (default: `0.775 kg/m` for 0.5" / 12.7 mm 7-wire strand). Used for all takeoff calculations.
- **Layer Color Overrides**: Assign specific AutoCAD ACI colors to Latitude (e.g. Magenta [6]) and Longitude (e.g. Cyan [4]) tendons for rapid field discrimination.

07 AutoCAD DWG/DXF Shop Drawing Export

7.1 Export Console Overview

Click the **"Export"** icon in the left navigation rail or press `Ctrl` + `E` to access the production drawing generator.

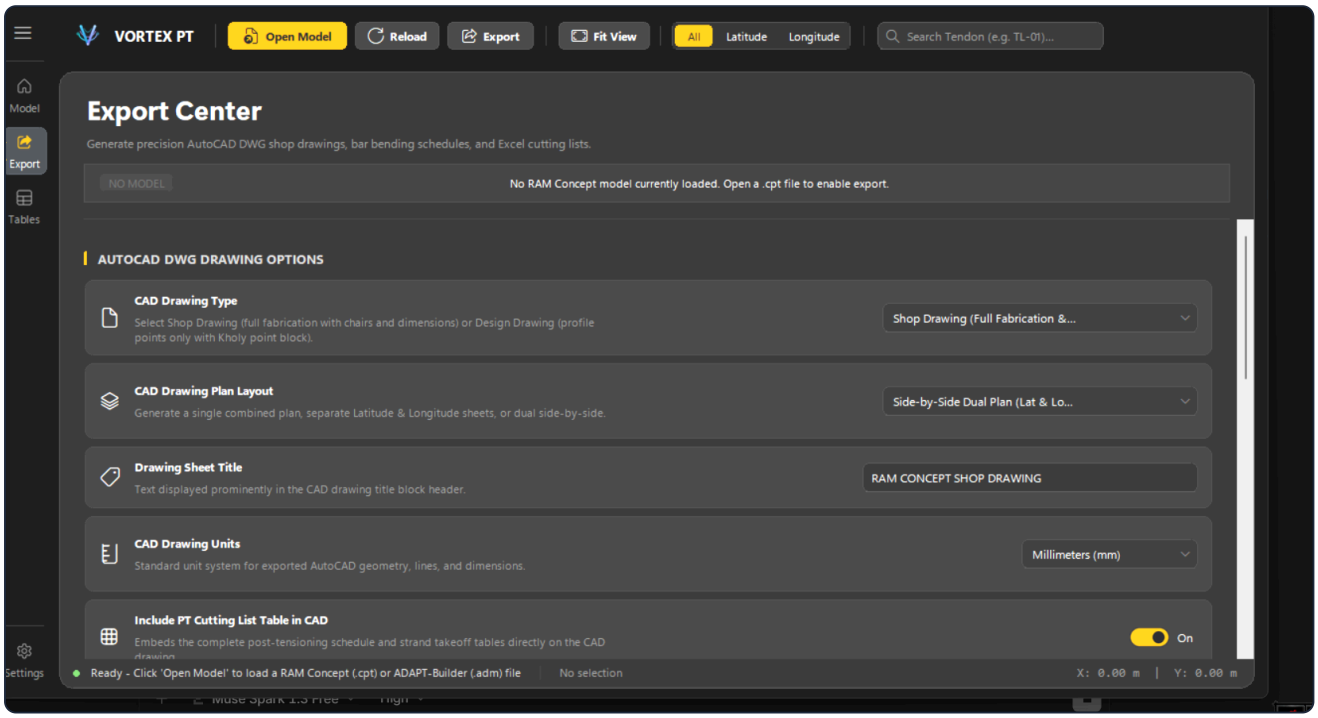


Figure 7.1: Production Drawing Export Console. Configure drawing types, layout styles, annotation layers, and direct AutoCAD DWG / DXF generation.

7.2 Production Drawing Configurations

A. CAD Drawing Type

- **Shop Drawing (Full Fabrication & Chairs)**: Complete fabrication-level drawing including chair support tags, remainder dimensions, tendon IDs, elongation markers, and anchor blocks.
- **Design Drawing (Profile Points Only)**: Streamlined consultant drawing with Kholy profile point blocks and high/low labels.

B. Sheet Layout Options

- **Side-by-Side Dual Plan [Recommended]**: Automatically offsets and places the Latitude drawing and Longitude drawing side-by-side on a single DWG layout for streamlined contractor printing.
- **Combined Plan**: Both Latitude and Longitude systems superimposed on a single floor plan.
- **2 Separate Plans**: Generates distinct independent files (e.g. `Model_Lat.dwg` and `Model_Long.dwg`).

7.3 Embedded Features

- **AutoCAD Layer Architecture**: All geometries are partitioned into dedicated layers (`PT-TENDON-LAT`, `PT-TENDON-LONG`, `PT-CHAIRS`, `PT-ANCHORS`, `PT-DIMS`) matching standard drafting protocols.
- **Embedded Schedule Table**: Optionally embeds the complete PT tendon schedule directly into model space.

7.4 Detailing Rules for Best Output in AutoCAD

1. **Recommended Sheet Layout**: Always select **Side-by-Side Dual Plan**. This places the Latitude and Longitude drawings side-by-side with an automatic bounding offset, creating clean, professional drawings ready for plotting.
2. **Dimension Style Separation**: Vortex PT generates separate dimension styles:
   - `PT_LONGITUDINAL_DIMS`: Tailored for tendon span lengths and remainder segments.
   - `PT_SPACING_DIMS`: Specially scaled for out-of-plan tendon spacing strings.
3. **Chair Deduction Tuning**: Verify high-point deduction (default: `10 mm`) and low-point deduction (default: `15 mm`) match site chair sizes. The minimum physical chair height is clamped to `20 mm` to prevent fabrication tickets for unbuildable chairs.

08 Fabrication Schedules & Takeoffs

8.1 Schedules Workspace

Click the **"Schedules"** icon in the left navigation rail to access interactive material cutting schedules and bar chair summaries.

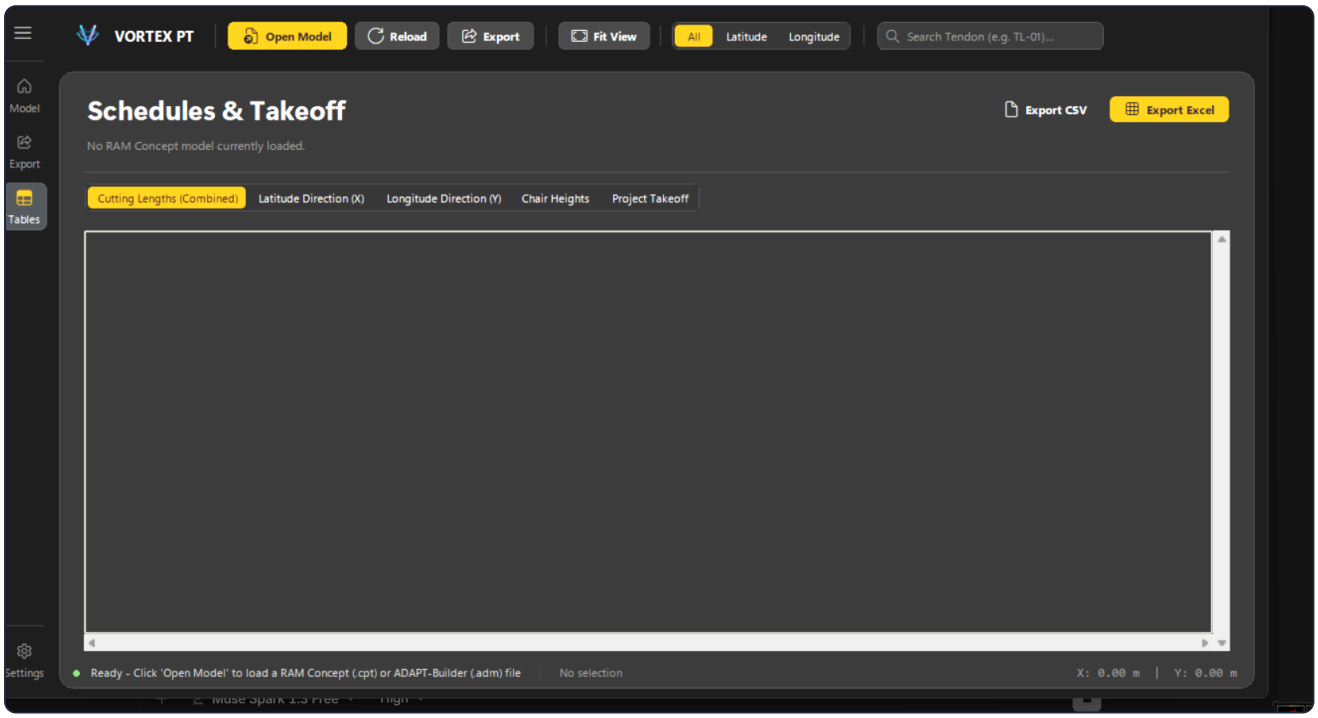


Figure 8.1: Interactive Fabrication Schedules. Displays tendon cutting schedules, strand lengths, theoretical elongations, and the Slab Chair Heights Summary Table.

8.2 Schedules Provided

1. **Combined Cutting Schedule**: Complete bill of materials listing Tendon Mark, Strand Count, Jacking Ends, Plan Length (m), Fabricated Cut Length (including stressing tail), Weight (kg), and Theoretical Elongation (mm).
2. **Directional Schedules (Latitude / Longitude)**: Filtered schedules partitioned for specific installation crews.
3. **Slab Chair Heights Summary Table**: Groups all chairs across the entire floor plate by fabricated height (e.g. 25 mm, 30 mm, ..., 200 mm) with exact counts for Latitude and Longitude.

8.3 Exporting to Microsoft Excel & CSV

Click the **"Export Excel (.xlsx)"** button or press `Ctrl` + `Shift` + `E`. Vortex PT generates a multi-tab workbook with pre-formatted engineering headers, mathematical column formulas, and print-ready page setups.

09 Hardware-Locked Cloud Licensing

9.1 Zero-Friction Online Activation

Vortex PT features a next-generation cloud licensing engine powered by a high-availability cloud database (Supabase 24/7). Customers no longer need to exchange hardware Machine IDs manually.

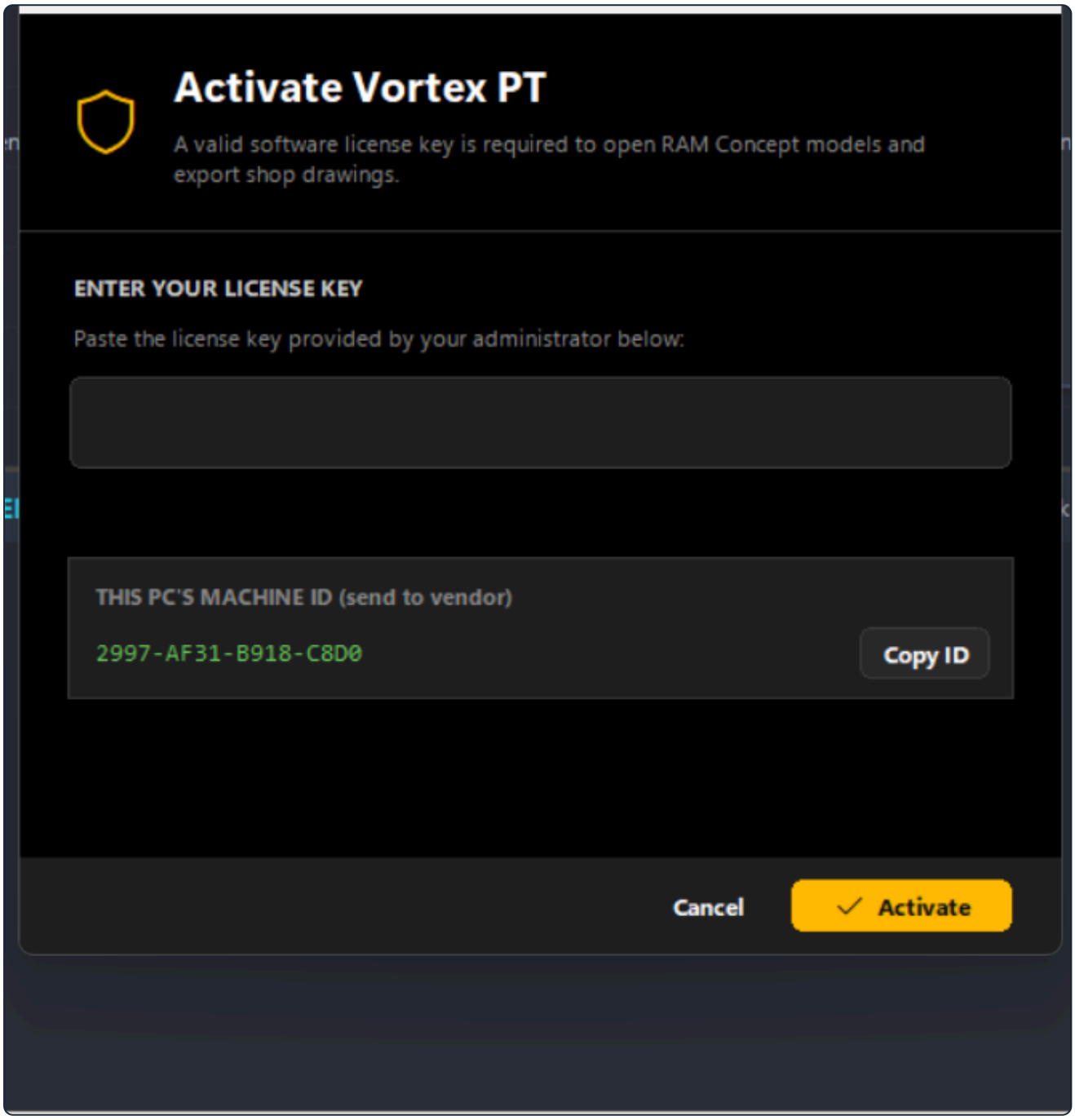


Figure 9.1: Instant Cloud Activation Dialog. Paste your license key and activate instantly without copying Machine IDs.

9.2 Activation & Workstation Binding Workflow

1. Upon launching the software, the Activation Dialog prompts for your license key.
2. Paste your issued license key (e.g. `VTX-CLOUD-XXXX-XXXX-XXXX`) into the input box and click **"Activate"**.
3. The application connects securely to the 24/7 cloud server, verifies your seat entitlement, and binds the license to your computer hardware.
4. **Permanent Hardware Locking**: For security and anti-piracy, the license is strictly bound to that specific workstation. Users cannot transfer or move the license.

9.3 1-Click Remote Renewals (No New Keys)

When your license subscription is extended or renewed by the software author:

- **Zero Re-keying**: You do **not** need to enter a new license key.
- **Instant Pickup**: In **Settings** → **Software License**, click **"Sync License"**. The software contacts the cloud server and updates your expiration date and days remaining immediately.
- **Background Auto-Sync**: Vortex PT periodically syncs license status in the background during normal operation.

9.4 30-Day Offline Grace Period & Air-Gapped Fallback

- **Site & Travel Freedom**: Once activated, a cryptographically signed lease receipt is stored securely in Windows DPAPI storage. You can work completely offline for up to **30 days** without internet connectivity.
- **100% Offline Air-Gapped Workstations**: For secure defense or high-security facilities with zero internet access, administrators can issue an offline machine-bound `VTX2-...` key that activates completely offline.

10 Keyboard Shortcuts & Engineering Reference

10.1 Complete Shortcut Reference

SHORTCUT	FUNCTION / COMMAND	CONTEXT
`Ctrl` + `O`	Open RAM Concept (.cpt) or ADAPT (.adm / .inb) Model	Global
`Ctrl` + `E`	Navigate directly to Export View	Global
`Ctrl` + `Shift` + `E`	Quick Export Schedules to Excel (.xlsx)	Global / Schedules
`Ctrl` + `F`	Focus Tendon Quick Search Field	Model View
`Ctrl` + `,`	Open Settings Console	Global
`Spacebar` / `F`	Fit Entire Floor Plate to Viewport Extents	Model View
`Esc`	Clear Tendon Selection & Close Floating Panels	Model View
`F5`	Reload & Re-parse Current Model File	Global

10.2 Standard 0.5" (12.7 mm) Strand Technical Specifications

PROPERTY	VALUE (SI METRIC)	DESIGN STANDARD
Nominal Strand Diameter	12.7 mm (0.50 inch)	ASTM A416 / ISO 6934
Nominal Cross-Sectional Area (Aps)	98.7 mm² (0.153 in²)	Grade 270 (1860 MPa) Low Relaxation
Strand Unit Weight	0.775 kg/m (0.521 lb/ft)	Standard 7-Wire High Tensile Strand
Modulus of Elasticity (Ep)	195,000 MPa (28,500 ksi)	Theoretical Elongation Calculations
Minimum Breaking Strength (Ppu)	183.7 kN (41.3 kips)	ASTM A416 Tensile Strength
Initial Jacking Force (0.80 fpu)	147.0 kN	Standard PTI Field Stressing

Technical Support & Licensing Inquiries

For software inquiries, license key renewals, or technical support, visit the official portal at **https://vortex-pt.com** or email **support@vortex-pt.com**.